Following this session, students will be able to:

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1. Interpret the pharmacokinetic properties of the antihypertensive agents in the context of their clinical significance.
2. Relate the mechanisms of action of the calcium channel blockers and beta blockers to their cardiovascular effects by giving examples of therapeutic applications.
3. Predict the adverse effects, drug interactions, and contraindications when given the mechanism of action and/or pharmacokinetic properties of the individual drug classes.
4. Select the appropriate antihypertensive agent and monitoring parameters for the individual patient based on the drug's pharmacologic effects when presented with a clinical case vignette.

CCBs: Pharmacokinetics (Class Properties)

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Good oral absorption but reduced bioavailability due to hepatic first-pass effect

High plasma protein binding; many are P-glycoprotein substrates

CYP-mediated metabolism, esp. CYP3A4

$t_{1/2}$ vary between 1.3 – 64 hours (drug-specific property)

- ***Extended release forms reduce dosing frequency of short-acting agents***

Some Pharmacokinetic Concepts Related to CCBs

Extended-release forms reduce the number of daily doses needed to maintain therapeutic drug levels.

Example:

- Nifedipine immediate release 3x daily (every 8 hours)
- Extended release 1x daily

Nifedipine extended-release appears to cause less reflex tachycardia than the immediate release formulation

Amlodipine has slow absorption, minimal peaks and troughs, and prolonged effect.

It causes less reflex tachycardia, possible due to lower peaks and long t½.

CCBs act from the inner side of the membrane.

The drugs bind more effectively **to open channels and inactivated channels**.

Similar to Na+ channel block by lidocaine

Drug binding reduces the frequency of opening in response to depolarization

↓

marked decrease in transmembrane calcium current

DHPs and non-DHPs

verapamil, diltiazem

Long-lasting arterial smooth muscle relaxation

Reduction in contractility throughout the heart

Decreases in SA node pacemaker rate and AV node conduction velocity

CCBs Therapeutic Uses and Effects

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Hypertension all CCBs initial or add-on therapy	**Hypertensive emergency** nicardipine, clevidipine (slow I.V. infusion)
relax smooth muscle of arterioles ↓ ↓ systemic vascular resistance ↓ ↓ BP	• DHP CCB *plus* labetalol or esmolol to prevent reflex tachycardia • Nicardipine: Potent, long-acting → duration ≤8 hours • Clevidipine: 1-minute t½ → duration 5-15 min

JNC8, black patients: CCB or thiazide diuretic recommended 1st-line
2025 ACC/AHA High Blood Pressure Guideline First-line therapy is the same for all patients.
Evidence supports individualized therapy based on comorbidities.

CCBs Therapeutic Uses and Effects

Angina
initial or add-on therapy

↓peripheral vascular resistance
→ ↓afterload
↓
↑coronary blood flow
↓
↑O_2 delivery to myocardium

Variant (vasospastic) angina:
CCBs induce coronary vasodilation.

Variant angina (coronary vasospasm)	• CCBs 1st-line • may combine with NTG
Exertional angina HR= heart rate SBP = systolic blood pressure	• BB *plus* DHP CCB more effective than either agent alone • Non-DHP act on the double product (HR x SBP): ↓HR x ↓SBP →↓ O_2 demand
Unstable angina (acute coronary syndrome)	• DHP CCB only when combined with BB • Non-DHP only for patients who do not tolerate BB, do not have left ventricular dysfunction, and show no signs of disturbed AV conduction

CCBs Additional Therapeutic Uses and Effects

Supraventricular tachyarrhythmias	Verapamil and diltiazem (hemodynamically stable patients) to slow automaticity and slow conduction through AV node, which may slow ventricular response.
Raynaud phenomenon	Long-acting DHP CCBs. At least moderate reduction in intensity of attacks and prevention of tissue loss is achievable in most patients.
Subarachnoid hemorrhage	Nimodipine (lipophilic, cerebral artery vasodilation) to reduce incidence and severity of ischemic deficits
Migraine	Verapamil may prevent migraine (weak and conflicting data)
Tocolytic	Nifedipine (for 48 hours only) to suppress premature contractions in preterm labor (24-34 weeks)

CCBs Adverse Effects

non-DHP	**DHP**
Bradycardia and worsening cardiac output (usually with I.V.) Constipation (especially verapamil)	Peripheral edema (most common side effect) Vasodilation → pre-capillary dilation and post-capillary reflex constriction → ↑ capillary pressure and permeability → redistribution of fluid into interstitium → edema
Both non-DHP and DHP: GERD aggravation	Immediate-release: • Headache, flushing, dizziness • Hypotension – reflex tachycardia (sympathetic activation)

Verapamil / Diltiazem Contraindications

Increased risk of AV block and/or severe depression of ventricular function:

- I.V. verapamil + I.V. beta blocker
- verapamil or diltiazem in patients with:
  - Ventricular dysfunction
  - SA or AV nodal conduction disturbances
  - Systolic blood pressures below 90 mm Hg

oral or particularly I.V.

Nifedipine (immediate-release) Contraindications

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pathic

Nifedipine immediate-release is not for use in patients with acute or unstable angina and STEMI

It may precipitate / aggravate myocardial infarction

- Causes abrupt vasodilation with reflex sympathetic activation
- MI may result from excessive hypotension, decreased coronary perfusion, or increased O_2 demand due to increased sympathetic tone and excessive tachycardia

Nifedipine short-acting formulations are not appropriate in the long-term treatment of angina or hypertension.

CCBs Drug Interactions

Verapamil + digoxin → digoxin toxicity

verapamil blocks Pgp →↑digoxin (Pgp substrate) levels →↑risk of digoxin toxicity

Pgp: P-glycoprotein

Verapamil + quinidine → excessive hypotension

- CYP3A4, Pgp inhibition, PD effects
- patients with idiopathic hypertrophic subaortic stenosis are at particular risk

CYP- and P-glycoprotein mediated interactions

Pharmacodynamic drug interactions

Beta blockers are recommended for hypertensive patients with concomitant heart disease and related comorbidities:

including

- Hypertension
  - Hypertensive emergencies: labetalol, esmolol (other classes of vasodilators are also used)
  - Intraoperative tachycardia, hypertension: *eg* propranolol, metoprolol, esmolol
- Ischemic heart disease (angina)
- Congestive heart failure
- Certain arrhythmias
- Hyperthyroidism
- Glaucoma (topical ophthalmic drops)

Labetalol is a recommended agent for the treatment of hypertensive emergency in pregnant persons.

Hypertension: Beta Blockers Mechanism of Action

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Hypertension is initiated and sustained at least in part by sympathetic neural activation.

Drugs, such as beta blockers, which block catecholamine actions, have important clinical effects.

- BBs reduce cardiac output and lower blood pressure in patients with hypertension (when taken regularly)
- Mechanisms probably include suppression of renin release and effects in the CNS

Hypertension: Beta Blockers Effects

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- *β* receptor blockade has relatively little effect on the *normal heart* of an individual at rest…
- …But, beta block has profound effects when sympathetic control of the heart is dominant, as during exercise or stress.
- BBs do *not usually* cause hypotension in healthy individuals with normal blood pressure.

Ischemic Heart Disease: Beta Blockers Mechanism of Action

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negative chronotropic effect
(particularly during exercise)

+

negative inotropic effect

+

reduction in
arterial blood pressure
(particularly systolic pressure)
during exercise

reduce myocardial O_2 consumption

at rest and during exertion

Ischemic Heart Disease: Beta Blockers Effects

In the treatment of exertional angina:

- BBs reduce myocardial O_2 consumption at rest and during exertion
  - also, they cause some increase in flow toward ischemic regions by increasing collateral circulation
- BBs have shown to:
  - Reduce the severity and frequency of attacks of exertional angina, and
  - Improve survival in patients who have had an MI

Ischemic Heart Disease: Beta Blockers Cautions

BBs' undesirable effects in angina include:

- ↑End-diastolic volume and
- ↑Ejection time

tend to increase myocardial O_2 requirement

Nitrates counterbalance these deleterious effects of BBs.

Congestive Heart Failure: BBs Mechanism of Action

Suggested mechanisms include:

1. Decrease unstable tachyarrhythmias
2. Attenuate sustained sympathetic activation
   - inhibit maladaptive proliferative cell signaling in the myocardium
   - reduce catecholamine-induced cardiomyocyte toxicity
   - decrease myocyte apoptosis
3. Reduced remodeling
   - decrease LV chamber size and increase LV ejection fraction
   - may decrease oxidative stress in the myocardium

Congestive Heart Failure: BBs Effects

Most patients with chronic heart failure respond favorably to certain β blockers.

- Caution: BBs can precipitate acute decompensation of cardiac function.

Only drugs shown to be effective in stable heart failure should be used.

- Studies with **bisoprolol, carvedilol, metoprolol** and **nebivolol** showed a reduction in mortality in patients with stable severe heart failure.
- (This effect was not seen with bucindolol.)

Arrhythmias: Beta Blockers Mechanism of Action

1. Reduce heart rate
2. Decrease intracellular Ca^{2+} overload
3. Inhibit afterdepolarization-mediated automaticity
4. Increase AV nodal conduction time (increased PR interval)
5. Prolong AV nodal refractoriness

Beta blockers are Vaughan-Williams class II antiarrhythmics

- ○ Some exert Na^{+} channel–blocking effects (membrane stabilizing effects) at high concentrations
  - Examples: propranolol, carvedilol

- ➢ BBs are useful in terminating re-entrant arrhythmias involving the AV node and in controlling ventricular response in atrial fibrillation or flutter.
- ➢ BBs can prevent recurrent infarction and sudden death in patients recovering from acute myocardial infarction.

Hyperthyroidism: Beta Blockers Effects

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Hyperthyroidism increases the basal metabolic rate and enhanced organ sensitivity to catecholamines

- It is associated with an increased number of β-adrenergic receptors.

BBs ameliorate symptoms:

- e.g. palpitations, tachycardia, tremulousness, anxiety, heat intolerance

Thyroid storm can cause supraventricular tachycardia and sudden death.

- clinical manifestations of thyrotoxicosis

BBs control:

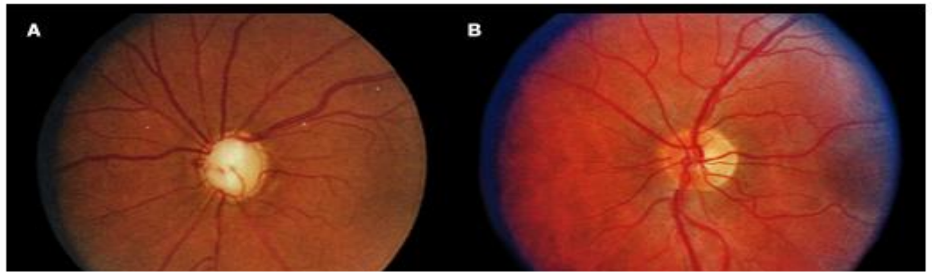
- pulse rate
- hypertension
- atrial fibrillation

Glaucoma: Ophthalmic Beta Blockers Mechanism of Action and Effects

- β receptors of the eye largely of the β_2 subtype
  - Located in the ciliary body epithelium and blood vessels
- BBs decrease IOP presumably by:
  - ↓ Production of aqueous humor, and/or
  - ↓ Ocular blood flow, which decreases the ultrafiltration responsible for aqueous production

IOP: intraocular pressure

Glaucoma



A) Extensive cupping of the optic disk typical of glaucoma
B) Compared to a normal optic disk

Courtesy of the W.K. Kellogg Eye Center, University of Michigan.
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- Little or no effect on pupil size or accommodation
- Do not cause blurred vision or night blindness
- Nasolacrimal drainage ⇒ **BBs potentially can cause adverse cardiovascular and pulmonary effects in susceptible patients.**

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Beta Blockers Adverse Effects and Mechanisms

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Excessive bradycardia	Block of cardiac β receptors (risk of bradyarrhythmias in patients with AV conduction defects)
Peripheral vascular insufficiency	Nonselective agents block β2-mediated peripheral vasodilation. (β1-selective agents are better tolerated in mild-moderate disease. But, caution in severe peripheral vascular disease or vasospastic disorders.)
Asthma / COPD exacerbation	Block β2-mediated bronchodilation → Airway resistance / Bronchospasm
Delayed recovery from insulin-induced hypoglycemia	1) Inhibit β2-mediated hepatic glycogenolysis / gluconeogenesis 2) Blunt perception of symptoms (tremor, tachycardia, nervousness) • β1-selective agents may be less prone to inhibit recovery from hypoglycemia

Beta Blockers Adverse Effects and Mechanisms

Decrease insulin sensitivity (non-vasodilating agents)	Mechanisms not understood • Can occur with both selective and nonselective agents
Increase plasma VLDL and decrease HDL	
Decrease release of free fatty acids from adipose tissue	Block β-mediated activation of hormone-sensitive lipase in fat cells (free fatty acids are an energy source for exercising muscle)
CNS effects	Fatigue; insomnia; nightmares (related to the drug's lipophilicity? No clear correlation has emerged.)
Decreased libido	Mechanism not understood

Beta Blockers Withdrawal Syndrome

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- Nervousness
- Tachycardia
- Increased blood pressure
- Increased intensity of angina
- Increased risk of sudden death

  MI reported in a few patients

- The withdrawal syndrome may involve upregulation or supersensitivity of β adrenoceptors
- **BBs should not be abruptly discontinued after prolonged regular use.**

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- Calcium channel blockers (CCBs) lower blood pressure by relaxing arteriolar smooth muscle and decreasing peripheral vascular resistance. Non-DHP CCBs Verapamil and diltiazem also decrease cardiac rate and force and reduce oxygen demand.
- CCBs are recommended first-line agents in the treatment of primary hypertension.
- Beta blockers lower blood pressure by decreasing cardiac output and cardiac contractility, and they reduce oxygen demand.
- Diuretics, ACEI, ARBs, CCBs, BBs, and alpha blockers are essentially equivalent in blood pressure lowering effect.
- Antihypertensives are frequently used in combination for increased efficacy.
- CCBs and beta blockers are used in the treatment of angina, supraventricular arrhythmias, and other uses.
- Side effects are related to the mechanisms.

Summary: Calcium Channel Blockers

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Non-DHP CCBs: Verapamil, Diltiazem

- Oral, IV, duration 4 to 24h (long-acting)
- Block L-type calcium channels in vessels and heart
- Reduce vascular resistance, cardiac rate, and cardiac force, which results in ↓ O_2 demand
- Used for angina prophylaxis, hypertension, rate control in atrial fibrillation, SVT treatment and prophylaxis
- Toxicity: AV block, acute heart failure; constipation; edema
- Interactions: Additive with other cardiac depressants and hypotensive drugs

DHP CCBs

- Oral, IV duration 4 to >24 h
- Block vascular L-type calcium channels to a greater extent than cardiac channels
- Reduce vascular resistance
  less cardiac depressant effect than non-DHP CCBs
- Prophylaxis of angina and treatment of hypertension

Immediate release nifedipine is *contraindicated.*

- Toxicity: Excessive hypotension, baroreceptor reflex tachycardia
- Interactions: Additive with other vasodilators

Summary of Beta Blockers

- Oral and parenteral; 4 to 24h (extended release)
- Beta blockers competitively block beta-adrenergic receptors. Some nonselectively block beta-1 and beta-2 receptors; others are relatively selective for beta-1 receptors. Some have additional properties, such as partial agonist effects, membrane stabilizing activity (high-dose), alpha-1 receptor block, and other effects.
- They decrease heart rate, cardiac output, myocardial O_2 demand.
- They are used for prophylaxis of angina, post-MI, compensated heart failure, as adjunctive agents in the treatment of hypertension, and other uses.
- Toxicity: Asthma, atrioventricular block, acute heart failure, sedation; beta-1selective blockers have less, but still significant, risk of bronchospasm
- Interactions: Additive with all cardiac depressants